



# Influence of El Niño southern oscillation on precipitation variability in Northeast Thailand



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## ARTICLE INFO

### Method name:

*Relationship between ENSO vs Climate Variability*

### Keywords:

Climate change

ENSO

Correlation

Wavelet transformation

Continuous wavelet transformation

Wavelet Coherence

## ABSTRACT

This study investigates the influence of El Niño Southern Oscillation (ENSO) on monthly precipitation anomaly (PPTA) in Northeast Thailand using 30 years (1993–2022) data obtained from 27 weather stations of the Thai Meteorological Department (TMD). Pearson correlation analysis elucidates the relationship between ENSO indices and PPTA. Results reveal significant correlations between ENSO indices and PPTA, providing insights into their intricate relationship. La Niña events have a stronger relationship with PPTA than El Niño events, with notable correlations observed.

- Strong La Niña phases, PPTA tends to increase across Northeast Thailand when Niño 3, 3.4, and 4 are more strongly negative -0.64, -0.50, and -0.65, respectively. These correlations provide valuable insights into the hydrological influence of ENSO in Thailand.
- After the ENSO event, there appears to be a 4–5 month lag period from the impact on PPTA that shows a higher correlation.
- Wavelet transform coherence (WTC) analysis dissects the temporal and frequency-specific nature of the relationship between ENSO and PPTA found during the frequency of 2–7 years. These findings underscore the importance of studying regional variations in ENSO impacts for PPTA in Northeast Thailand.

## Specifications table

|                                        |                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------|
| Subject area:                          | Engineering                                                                                           |
| More specific subject area:            | Modeling and Forecasting                                                                              |
| Name of your method:                   | Relationship between ENSO vs Climate Variability                                                      |
| Name and reference of original method: | NA.                                                                                                   |
| Resource availability:                 | Data used to support the study's findings can be obtained from the corresponding author upon request. |

## Background

El Niño Southern Oscillation (ENSO) was first identified by Sir Gilbert Thomas Walker in 1924 [1] during his meteorological study, focusing on wind distribution and sea surface temperatures (SST) in storm theory [2]. ENSO consists of El Niño, La Niña, and Neutral phase, each characterized by different oceanic and atmospheric conditions. Neutral conditions, or “ENSO-neutral,” occur when the

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oceanic and atmospheric patterns in the tropical Pacific are neither in El Niño nor La Niña. During these periods, the SST in the central and eastern equatorial Pacific Ocean are close to their long-term average [3], and the trade winds and atmospheric pressure patterns in the Pacific Ocean are near normal caused by the Walker Circulation, which involves trade winds blowing from east to west across the equatorial Pacific, remains near its average strength. There is no weakening or strengthening of the trade winds [4]. During an El Niño event, SST in the central and eastern equatorial Pacific Ocean are warmer than average, while La Niña is the opposite of El Niño. During La Niña events, SSTs in the central and eastern equatorial Pacific are cooler than average. Each phase varies in intensity and is categorized as weak, medium, and strong [3,5,6], contributing valuable insights into global climate dynamics [7]. ENSO events influence global climate patterns, influencing precipitation, temperature, and weather extremes worldwide. The impacts of El Niño and La Niña vary from region to region [8]. Kirtphaiboon et al. (2014) studied rainfall anomalies in the La Niña and El Niño phases. Weak-ENSO periods experience higher rainfall anomalies than strong-ENSO periods due to their frequency. La Niña caused the increase in rainfall in Southeast Asia, while El Niño led to reduced rainfall [9]. During El Niño, autumn rainfall in the Extended Central Vietnam (ECV) area declines by 10 to 30 %, attributed to weakened Northeast monsoon circulation and anticyclonic anomalies over the East China Sea. Conversely, La Niña boosts ECV's autumn rainfall by 9 to 19 % through strengthened Northeast monsoon and increased moisture supply from the Central and Western Pacific. La Niña also shifts rainfall distribution, especially in October and November, while El Niño shows spatial rainfall deficits in September and October, mainly due to weakened monsoon and reduced moisture supply [10].

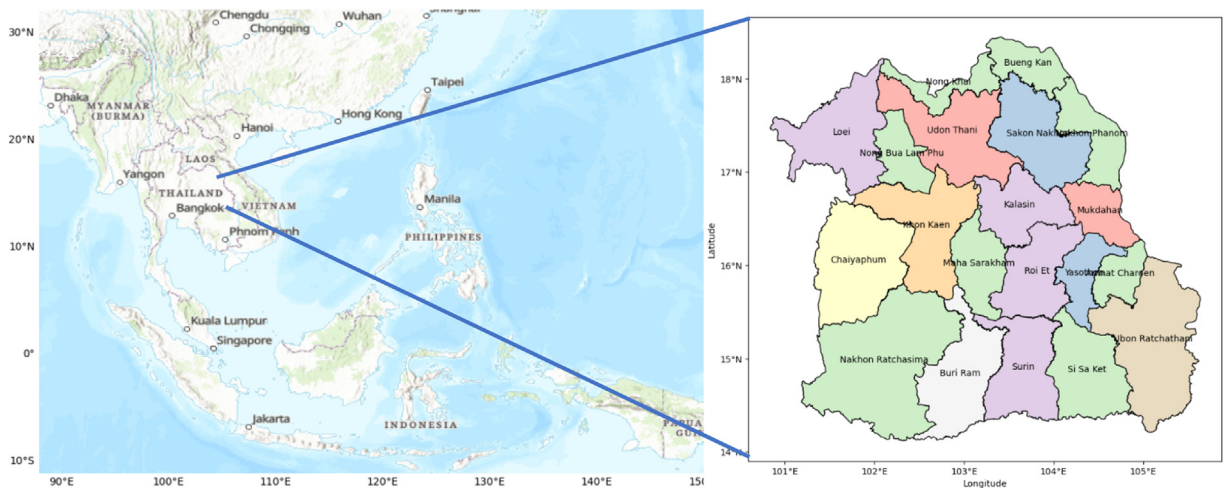
In Thailand, El Niño tends to bring reduced rainfall, which can lead to drought conditions and water shortages. It can also increase the risk of forest fires due to the dry conditions [11]. ENSO dynamics pose a dual risk: during El Niño, water scarcity is often made worse by the El Niño phase elevated drought, while La Niña can reduce drought impact but may lead to hazards due to excessive rainfall because of increased rainfall in monsoon [12,13]. Muangsong et al. (2020) found ENSO's substantial impact on northwest Thailand's monsoon rainfall, with ENSO cycles notably influencing drought occurrences since 1980 via changes in the Walker circulation [14]. Climate variability, mainly standardized precipitation evapotranspiration index (SPEI), significantly affects Thailand's agriculture, with wet La Niña conditions contrasting dry El Niño periods, underlining ENSO's crucial role in the country's agricultural sector [15]. The effects of ENSO events on Thailand's climate are subject to variation based on the event's strength, duration, and regional climate factors [16]. Every ENSO event involves a two to eight-year cycle [17]. The precise impact of ENSO remains elusive, prompting considerable interest in studying its mechanisms and patterns. Pearson's correlation and wavelet analysis emerge as vital tools for discerning the frequency of signals from ENSO events to capture both the overall relationships and the nuanced, time-varying patterns in ENSO signals, offering an approach to analyzing complex climate data. Pearson's correlation is a statistical tool that measures the linear relationship between two variables. It helps identify whether there is a consistent relationship between ENSO signals and precipitation (PPT) over time. Wavelet analysis can identify oscillations with phase and amplitude modulation, such as recognition of ENSO indices. It has advantages over the Fourier transform in application patterns [18,19]. For example, Torrence and Compo (2011) investigated changes in ENSO variance on interdecadal timescales, finding higher power in the Southern Oscillation Index (SOI) and Niño 3 (sea surface temperature anomaly (SSTA)) during 1880–1920 and 1960–1990, with lower power during 1920–60, suggesting a possible 15-year modulation of variance. Their study affirms wavelet analysis as a valuable tool for examining localized variations in time series, elucidating dominant modes of variability [20]. Tamaddun et al. (2017) also explored the relationship between streamflow in the western United States and oceanic-atmospheric indices Pacific Decadal Oscillation (PDO) and ENSO. Employing continuous wavelet transform (CWT) and Wavelet Transform Coherence (WTC), they found concurrent variations between streamflow and both indices, with PDO showing high correlation in bands of 8–10 years and beyond 16 years, while ENSO demonstrated significant correlation with streamflow in a 10–12-year band [21].

This study investigates the relationship between ENSO and precipitation anomaly (PPTA) in Northeast Thailand, utilizing Pearson's correlation analysis to quantify relationships between ENSO indices and PPTA for 1993–2022. This statistical approach lays the groundwork for identifying initial connections and potential areas of influence. Additionally, WTC is applied to explore how the periodicity of ENSO impacts varies over time across different frequency bands. WTC enables the examination of the temporal recurrence of ENSO-related signals in PPTA, enhancing understanding of their transient nature. This method facilitates the detection of significant co-varying patterns and time-dependent relationships that traditional correlation measures may obscure. The research structure encompasses the study area, dataset acquisition, data preprocessing, methodology, results, discussion, and conclusion, organized sequentially.

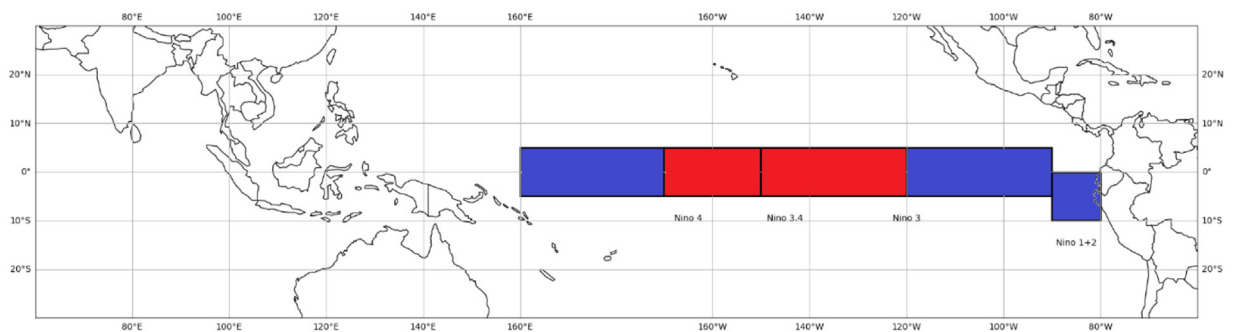
## Method details

### Study area

Northeast Thailand lies between 14°N and 18°N and longitudes 101°E and 105°E. Northeast Thailand (Fig. 1), or Isan is a geographically distinct region characterized by its plateau landscape known as the Korat plateau [15,22], which the Mekong River covers from north to east and the Phetchabun Mountains to the west. The region spans approximately 170,000 km<sup>2</sup>. The climate is typically tropical savanna, with a marked seasonality in precipitation due to the influence of the monsoon. Precipitation in Northeast Thailand is heavily influenced by the monsoon season. The annual rainfall in the region varies from about 1100 to 1900 mm, with the heaviest rains typically occurring in August and September [23]. The tropical climate has an average PPT of 1470 mm, concentrated in the rainy season from mid-May to mid-October and a maximum between August and September [24]. During this period, the region receives most of its annual rainfall, which is essential for its primarily agricultural economy and includes rice, rubber, and sugarcane [25]. The wet season brings frequent and sometimes heavy rains, contributing to the rice paddies and other crops. Conversely, the



**Fig. 1.** Southeast Asia and Northeast Thailand (study area).



**Fig. 2.** Niño regions over the Pacific Ocean.

dry season, which extends from November to April, sees significantly reduced rainfall, leading to arid conditions and a reliance on irrigation for farming.

#### Data acquisition

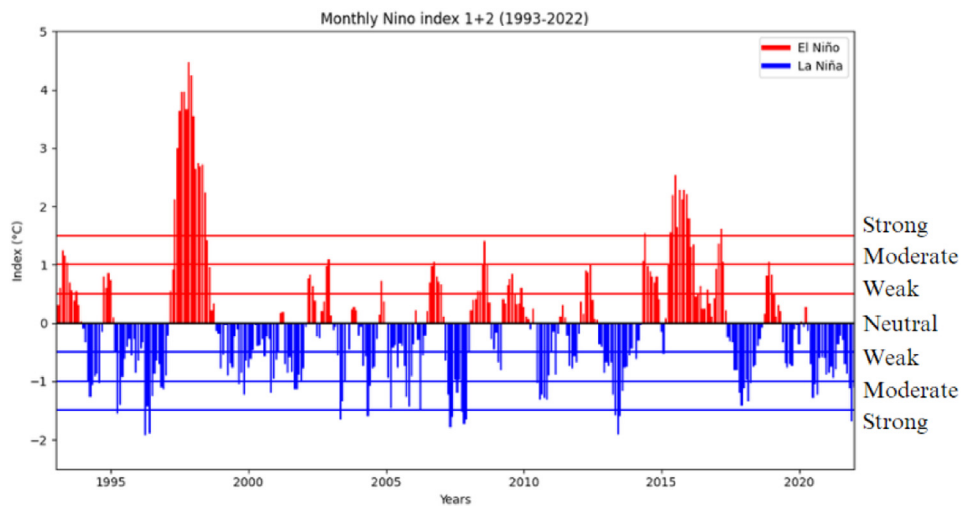
This study utilized monthly PPTs from the Thai Meteorological Department (TMD) from 27 weather stations covering 30 years (1993–2022). The analysis integrated PPT obtained from TMD alongside the Niño indices sourced from NOAA (ERSST.v5 dataset) [26], assisting in a comprehensive analysis of ENSO's impact on Northeast Thailand. ENSO indices, crucial for identifying the intensity and duration of El Niño and La Niña phases, played a pivotal role in understanding the influence of ENSO events on PPT [27]. The regional areas are shown in Fig. 2, Niño 1 + 2 (0–10S, 90W–80 W): this is the easternmost and smallest of the Niño regions, corresponding to the coastal South American region where the locals first noticed El Niño. Niño 1 + 2 highest variance is often found [28]. Next, Niño 3 (5N–5S, 150W–90 W): after discovering that the critical area for coupled ocean-atmosphere interactions for ENSO is located further west, scientists shifted their primary attention from monitoring and forecasting El Niño events to Niño region 3.4. Niño 3.4 (5N–5S, 170W–120 W) can be understood as the average equatorial SSTs over the Pacific. Lastly, Niño 4 (5N–5S, 160E–150 W): Sea Surface Temperature Anomaly (SSTA) in the center equatorial Pacific are captured by this index. Compared to the other Niño regions, with maximum warm SSTA located primarily in this region [29]. Therefore, the Niño index for each Niño region can be calculated, and Fig. 3 displays the time series of the Niño indices, consequently.

#### Data preprocessing

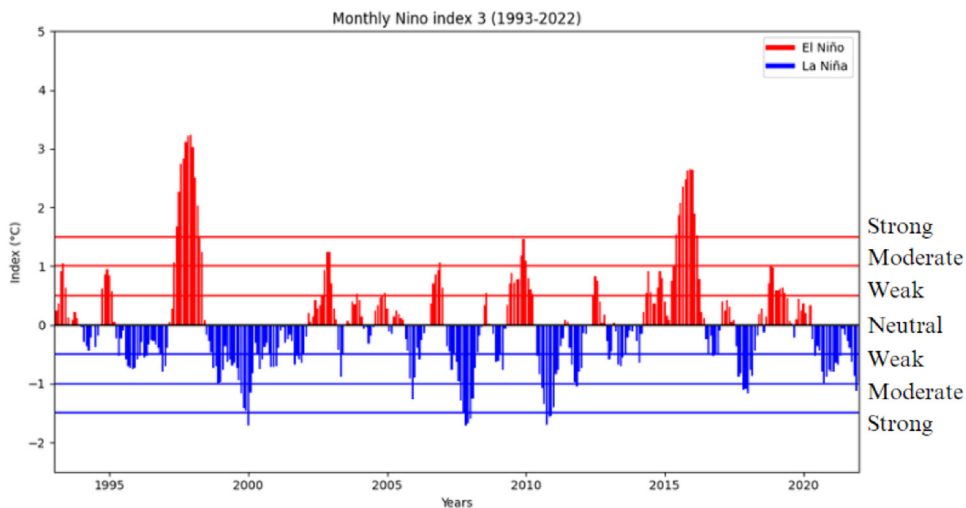
Data handling and preprocessing are essential steps in the data analysis process. It is common for collected data to exhibit missing values, errors, or inconsistencies [30]. To handle missing values, the Multi Linear Regression (MLR) proposed by Wangwongchai et al. (2023) was used to impute missing values in PPT [31]. Integrating various PPT and ENSO indices sources into an organized data set is accepted to help comprehensive analysis, including statistical analysis. This process ensures the appropriate alignment of timestamps and data formats. Statistical analysis involves collecting, organizing, interpreting, and presenting data to make meaningful conclusions and decisions. It involves applying various statistical techniques and methods to data to uncover patterns, relationships, and trends

[32]. Therefore, statistical analysis employs summary coefficients to encapsulate a dataset representing the entire population or a sample. Descriptive statistics are categorized into measures of central tendency and variability. Central tendency measures encompass the mean, median, and mode, while variability measures include Standard Deviation (SD), mean, minimum, and maximum values. This subsection provided descriptive statistics for PPT. Monthly PPT from TMD consists of meteorological stations, with 27 stations in Northeast Thailand.

In a period of 30 years, from 1993 to 2022, it is a total of 360 months. The problem that is always encountered when using data is data loss, which occurs on certain days and months. As for the missing data, Multi Linear Regression (MLR) was used to fill the lost PPT by using data from nearby stations to estimate the missing data, which gave good results [31]. However, the goal is to use as much of the collected information as possible while also considering the appropriate statistical techniques for analysis. Therefore, data handling involves addressing missing data and selecting appropriate statistical methods to analyze and report the collected data [33,34]. Details of the descriptive statistics of the PPT data at each station are shown in Table 1.

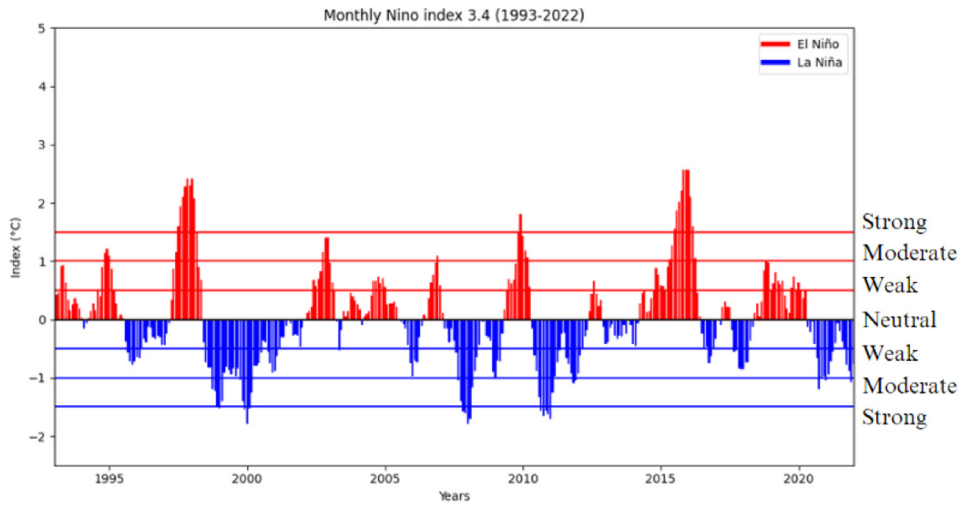


(a)

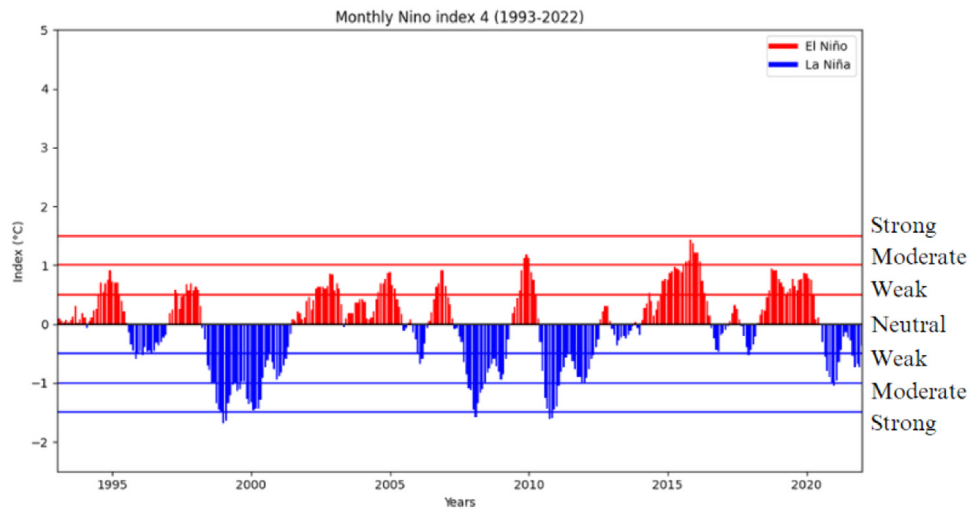


(b)

**Fig. 3.** Nino index over January 1993 to December 2022 (a) Niño 1 + 2, (b) Niño 3, (c) Niño 3.4 and (d) Niño 4.



(c)



(d)

Fig. 3. Continued

### The Pearson correlation

The Pearson correlation coefficient ( $r$ ) is a descriptive statistic, meaning that it summarizes the characteristics of a dataset by finding a relationship between two variables. Specifically, it describes the strength and direction of the linear relationship between two quantitative variables, shown in Eq. (1). The value of Pearson correlation is a number between  $-1$  and  $1$  that measures the strength and direction of the relationship. If the value approaches one, two variables have a direct relationship, and the opposite for  $-1$ . If the value approaches 0, the two variables do not have a relationship [35].

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}, \quad (1)$$

where,

$x_i, y_i$  are the actual value at time  $i$ ,  
 $\bar{x}, \bar{y}$  are the mean values.

**Table 1**  
Descriptive monthly PPT (mm) statistics in Northeast Thailand over 1993–2022.

| Station no. | min | max    | mean  | std   |
|-------------|-----|--------|-------|-------|
| 352201      | 0.0 | 688.5  | 154.4 | 155.9 |
| 353201      | 0.0 | 532.6  | 108.2 | 106.8 |
| 353301      | 0.0 | 510.1  | 107.3 | 102.8 |
| 354201      | 0.0 | 709.5  | 123.7 | 126.6 |
| 356201      | 0.0 | 798.8  | 140.5 | 144.5 |
| 356301      | 0.0 | 796.3  | 133.0 | 137.6 |
| 357201      | 0.0 | 1055.5 | 194.3 | 226.8 |
| 357301      | 0.0 | 1141.5 | 170.0 | 193.9 |
| 381201      | 0.0 | 460.3  | 107.7 | 103.7 |
| 381301      | 0.0 | 569.4  | 104.0 | 102.2 |
| 383201      | 0.0 | 803.3  | 127.9 | 142.8 |
| 387401      | 0.0 | 577.2  | 112.4 | 115.6 |
| 388401      | 0.0 | 561.8  | 102.6 | 118.2 |
| 403201      | 0.0 | 516.1  | 99.6  | 101.0 |
| 405201      | 0.0 | 661.1  | 115.2 | 119.8 |
| 405301      | 0.0 | 789.2  | 114.8 | 117.7 |
| 407301      | 0.0 | 749.1  | 142.8 | 149.4 |
| 407501      | 0.0 | 661.2  | 144.8 | 149.9 |
| 409301      | 0.0 | 827.5  | 127.1 | 137.4 |
| 431201      | 0.0 | 546.1  | 96.5  | 94.8  |
| 431301      | 0.0 | 435.0  | 108.0 | 92.5  |
| 431401      | 0.0 | 428.2  | 95.6  | 88.5  |
| 432201      | 0.0 | 676.3  | 129.1 | 124.0 |
| 432301      | 0.0 | 651.5  | 129.1 | 124.2 |
| 432401      | 0.0 | 549.6  | 115.2 | 112.8 |
| 436201      | 0.0 | 521.5  | 91.3  | 97.9  |
| 436401      | 0.0 | 419.2  | 113.4 | 98.9  |

### Wavelet transformation

Wavelet transform is a mathematical tool used for simultaneously analyzing signals and data in both time and frequency domains [36]. The basic idea behind wavelet transform is to decompose a signal into different frequency components at different scales [37]. WCT is achieved by convolving the signal with a scaled and translated version of a reference wavelet function. The wavelet function is a small wave-like oscillation localized in both time and frequency, allowing it to capture details at different scales. There are two main types of wavelet transforms are Continuous Wavelet Transform (CWT) and Discrete Wavelet Transform (DWT) [38,39]. For CWT, the wavelet function is continuously scaled and translated across the entire signal, providing a continuous-time and continuous-frequency representation. This study uses wavelet transform to analyze climate data and ENSO indices and identify patterns, trends, and periodicities at different temporal scales. Additionally, wavelet transform is employed in conjunction with other statistical techniques, such as coherence analysis, to investigate complex relationships between PPTA and ENSO events [40,41]. The CWT formula is given by

$$W_s(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} s(t) \psi^* \left( \frac{t-b}{a} \right) dt, \quad (2)$$

where  $a$  and  $b$  are the scales and time shifts of a reference wavelet  $\psi$ , respectively,  $\psi^*$  and are the complex conjugate of the reference wavelet, and they are time and the timescale representation of the signal. The formula of CWT for the discrete sequence  $x_n$  is defined with a scaled and translated version of  $\psi_0(\eta)$  [42] given by

$$W_n^x(s) = \sqrt{\frac{\delta t}{s}} \sum_{n'=1}^N x_{n'} \psi^* \left( \frac{(n' - n) \delta t}{s} \right), \quad (3)$$

where  $N$  is the number of records and  $\delta_t$  is the sampling interval. The reference wavelet is also commonly known as the mother wavelet, and an essential feature of the mother wavelet [18] is the following

$$\psi(t) = \frac{1}{\sqrt{a}} \psi \left( \frac{t-b}{a} \right), \quad (4)$$

when employing wavelets for feature extraction, one popular mother wavelet is the Morlet since it presents a good balance between temporal and frequency localization [43]. The formula of the Morlet wavelet is given by

$$\psi_0(\eta) = \frac{1}{\sqrt[4]{\pi}} e^{i\omega_0 \eta} e^{-\frac{1}{2} \eta^2}, \quad (5)$$

$\omega_0 = 6$  to satisfy the admissibility condition [36] defining the number of oscillations, the wavelet has a central frequency of around 6 radians per unit time.

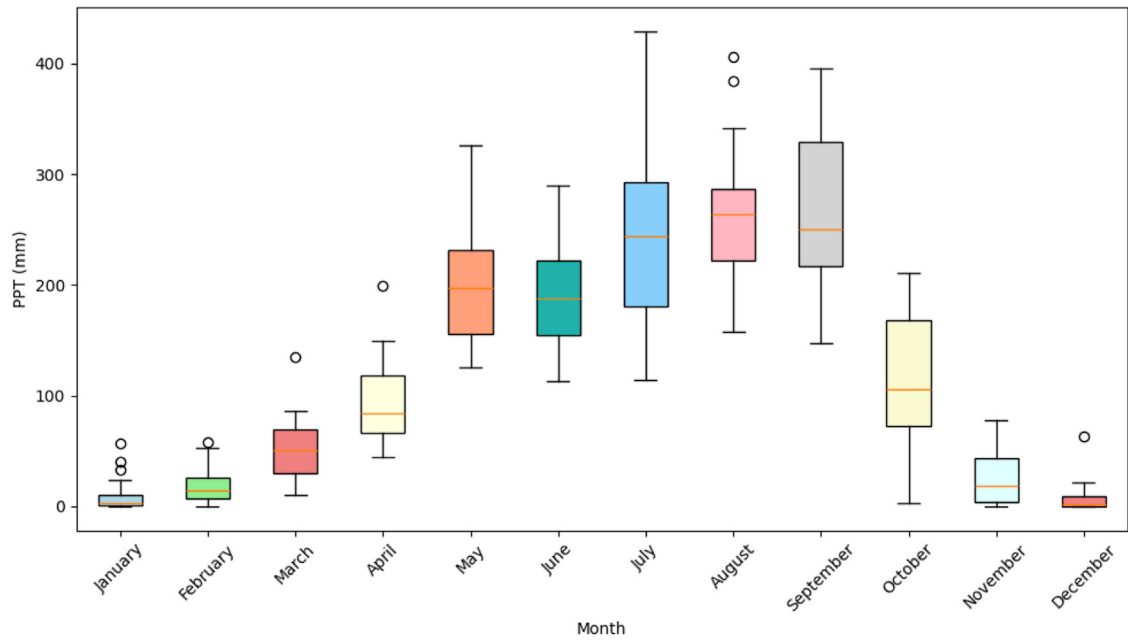


Fig. 4. Distribution of PPT for each month in Northeast Thailand over 1993–2022.

Wavelet coherence measures the local correlation between two-time series in both time and frequency domains. It is defined as the ratio of the cross-spectrum to the product of the spectrum of each series. Wavelet coherence provides information about the strength and consistency of the relationship between two variables across different frequencies and time intervals. Smoothing is necessary to avoid identically 1 coherence values, and it is achieved through convolution with a Gaussian in time and a rectangular window in scale. Theoretical distributions for wavelet coherence have not been derived yet, so statistical significance is typically assessed using Monte Carlo simulation methods [44]. In the context of climate time series analysis, cross-wavelet tools have been used to show how the relationship between PPTA and Climate indices, such as the ENSO index, has changed and evolved, with variations observed across different frequencies. The formula of wavelet coherence is given by

$$R_n^2(s) = \frac{\left| S(s^{-1} W_n^x(s) W_n^y * (s)) \right|^2}{S(s^{-1} |W_n^x(s)|^2) S(s^{-1} |W_n^y * (s)|^2)}, \quad (6)$$

where  $S(X) = S_{scale}(S_{time}(W(a, b)))$  a smoothing operator consists of  $S_{scale}$  denoting smoothing along the wavelet scale axis and  $S_{time}$  denoting smoothing in time. The following is a suitable smoothing operator for the Morlet wavelet:

$$S_{time}(X)|_s = \left( X_n(s) \times c_1^{-\frac{t^2}{2s^2}} \right),$$

$$S_{scale}(X)|_n = \left( X_n(s) \times c_2 \prod (0.6s) \right)|_n \quad (7)$$

Furthermore, the Monte Carlo method was applied in this study to conduct the significance test of the wavelet coherence spectrum. Only the arrows  $R_n^2 > 0.7$  are marked in the wavelet coherence spectrum, and the mean phase angle and 95 % confidence interval are drawn for the wavelet coherence spectrum [45].

## Method validation

Pearson's correlation analysis presents the correlation coefficients between Niño indices and PPTA in Northeast Thailand with lag time and different severity. In addition, WTC plots depict significant coherence levels between ENSO indices and PPTA. Highlight coherent phases and time-dependent relationships revealed by wavelet analysis.

Fig. 4 shows the monthly distribution of PPT in Northeast Thailand. Each box represents the interquartile range of PPT values for each month, depending on 1993–2022, with the median indicated by a horizontal line within the box. The whiskers extend to the minimum and maximum values within 1.5 times the interquartile range, while the dots represent outliers. The mean monthly PPT for each month was discovered to vary. It reveals that the wettest months start in May and end in October (rainy), with significant variability and some high outliers. The PPT decreases notably from October onward, reaching its minimum in December and January (winter) and increasing slowly from February to May (summer). Fig. 4 indicates a distinct rainy season peaking around July, August,



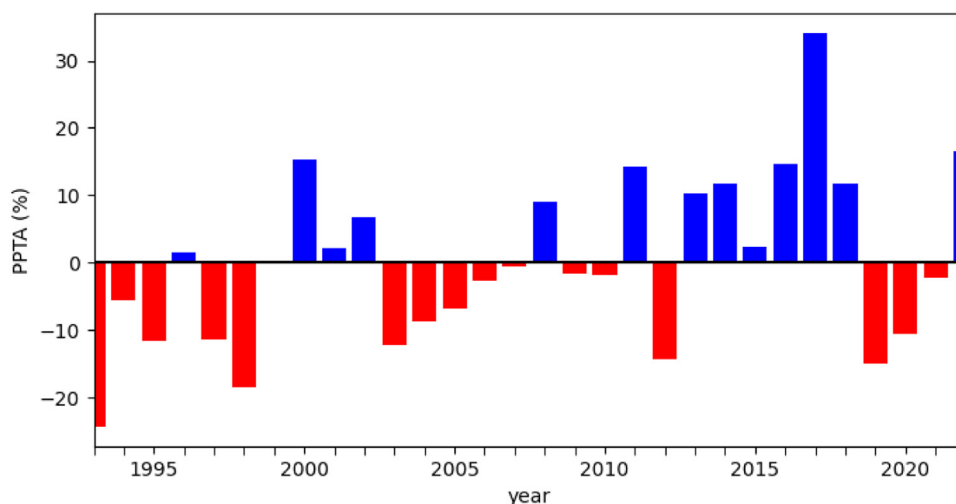


Fig. 5. Annual PPTA (percentage) in Northeast Thailand over 1993–2022 (mean 1470.80 mm).

Table 2

Pearson's correlation between Niño indices and Monthly PPTA with time lag (0 – 12 months).

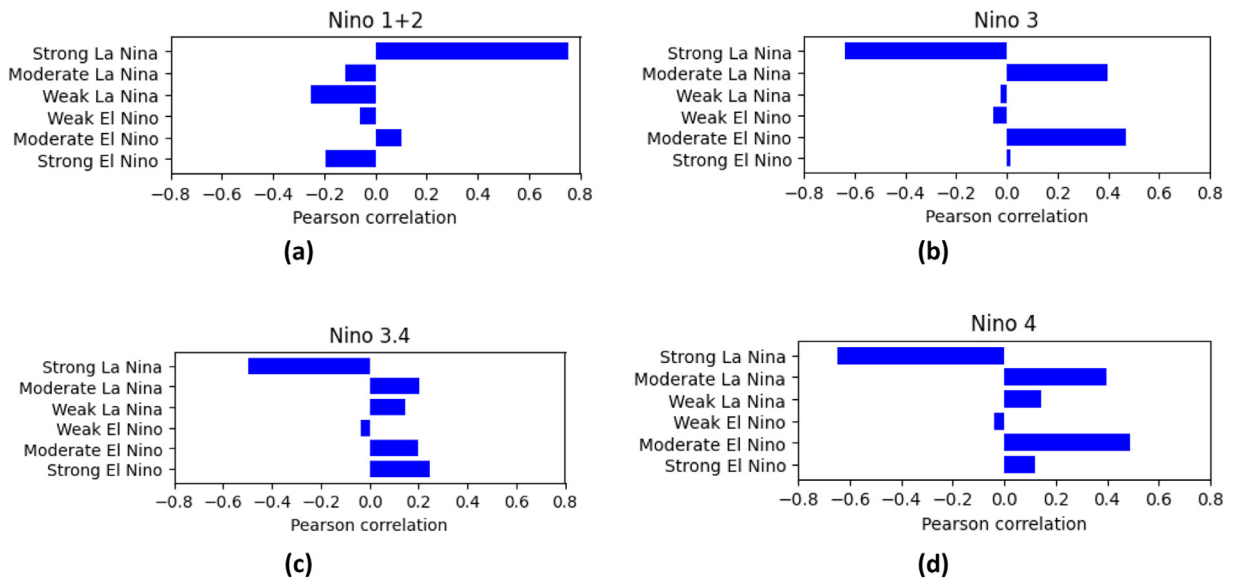
| Lag PPTA | Niño 1 + 2 |          | Niño 3     |          | Niño 3.4   |          | Niño 4     |          |
|----------|------------|----------|------------|----------|------------|----------|------------|----------|
|          | r          | P-values | r          | P-values | r          | P-values | r          | P-values |
| 0        | -0.08      | 0.15     | -0.05      | 0.31     | -0.11 (*)  | 0.04     | -0.08      | 0.13     |
| 1        | -0.05      | 0.33     | -0.05      | 0.37     | -0.10      | 0.07     | -0.08      | 0.12     |
| 2        | -0.03      | 0.55     | -0.07      | 0.21     | -0.12 (*)  | 0.03     | -0.09      | 0.08     |
| 3        | -0.07      | 0.19     | -0.09      | 0.10     | -0.12 (*)  | 0.02     | -0.12 (*)  | 0.02     |
| 4        | -0.06      | 0.25     | -0.13 (*)  | 0.01     | -0.14 (**) | 0.01     | -0.16 (**) | 0.00     |
| 5        | -0.09      | 0.09     | -0.14 (**) | 0.01     | -0.12 (*)  | 0.02     | -0.15 (**) | 0.00     |
| 6        | -0.04      | 0.44     | -0.11 (*)  | 0.03     | -0.10      | 0.06     | -0.13 (*)  | 0.01     |
| 7        | -0.04      | 0.41     | -0.14 (*)  | 0.01     | -0.08      | 0.12     | -0.13 (*)  | 0.02     |
| 8        | -0.03      | 0.53     | -0.09      | 0.11     | -0.08      | 0.13     | -0.10      | 0.05     |
| 9        | -0.06      | 0.23     | -0.08      | 0.11     | -0.06      | 0.26     | -0.09      | 0.08     |
| 10       | -0.05      | 0.31     | -0.10      | 0.07     | -0.06      | 0.25     | -0.09      | 0.09     |
| 11       | -0.07      | 0.16     | -0.08      | 0.12     | -0.06      | 0.28     | -0.09      | 0.11     |
| 12       | -0.06      | 0.27     | -0.07      | 0.21     | -0.05      | 0.35     | -0.06      | 0.22     |

and September, followed by a decrease during winter and slowly increasing in summer. The monthly PPT appears to follow a seasonal pattern. Fig. 5 shows the annual PPTA for higher/lower than mean annual PPT over 1993–2022, in which annual PPTA lower than the mean value (1470.8 mm) was as follows: 1993–1995, 1997–1999, 2003–2007, 2009–2010, 2012, and 2019–2021. The annual PPT with a higher than mean value was as follows: 1996, 2000–2002, 2008, 2011, 2013–2018, and 2022. 2017 was the year with the most PPT over 1993–2022, with a peak PPT of 172.8 mm, more than the mean of 502.0 mm (i.e., 34.13 %). The lowest PPT, roughly 1112.3 mm, was found in 1993, less than the mean PPT by 358.5 mm (–24.38 %).

#### Pearson's correlation analysis

The correlation between Niño indices and PPTA at various lags is shown in Table 2. Each row indicates a different time lag (0 to 12 months), indicating the impact on the PPTA after changes in the Niño indices. The negative correlation tends to get stronger as the lag lengthens. The time lag in correlations suggests that the impact of SSTa in the Niño regions on PPTA may be delayed due to the propagation of oceanic and atmospheric waves, which can take several months to influence distant regions. At lag 0, all regions show negative correlations with PPTA, indicating that at the same time, increases in Niño indices are associated with decreases in PPTA and true for the opposite. As the lag increases, the correlation values vary, indicating the temporal relationship between SSTa in the Niño regions and PPTA in Northeast Thailand. The following shows the statistical significance by critical value for the correlation coefficient at a given significance level based on the t-distribution and p-values of the Niño indices-PPTA relationship, which (\*) represents a 5% significance level and (\*\*) represents a 1% significance level. Niño 3, a lagged relationship from 4 to 7 months lag, is shown at a statistical significance of 5%. At a 5-month lag, the significance is even stronger at 1%. For Niño 3.4, a lag in the relationship at the same month and 2–5 months lag, with a statistical significance of 5% and 1% at 4 months lag. Niño 4 demonstrates a lag connection from 3 to 7 months lag at a statistical significance of 5% and 4 to 5 months lag. The significance is stronger at 1%. For Niño index 1 + 2 and 3, the peak negative correlation occurs at lag 5 (–0.09 and –0.14). For Niño index 3.4





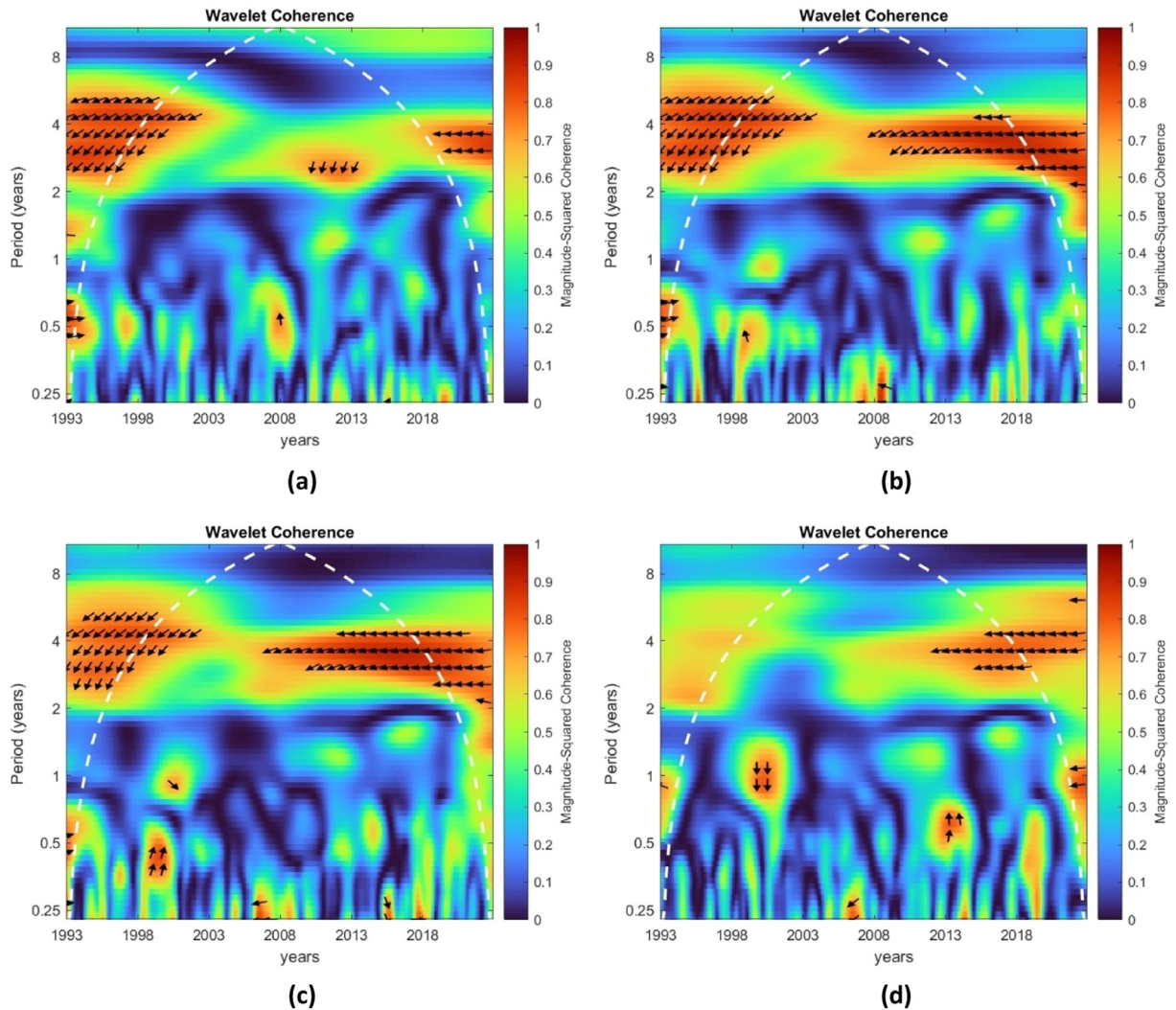
**Fig. 6.** Pearson's correlation between Niño index (a) Niño 1 + 2, (b) Niño 3, (c) Niño 3.4, (d) Niño 4 and PPTA of weak, medium, and strong for El Niño/La Niña.

and 4, the strongest negative correlation occurs at lag 4 ( $-0.14$  and  $-0.16$ ). So, when SSTa in the Niño regions (Niño indices) are high, the PPTA tends to be lower, and it is also true for the opposite that is, while El Niño happens, PPTA over Northeast Thailand decreases from normal, and when La Niña happens, PPTA over Northeast Thailand increases. The correlations for Niño 1 + 2, Niño 3, Niño 3.4, and Niño 4 show fluctuating patterns over the 12 months, indicating that the influence of SSTa in these regions on PPTA is not linear and may be affected by other climatic factors.

The Niño (1 + 2, 3, 3.4, and 4) exhibits various correlations with PPT anomalies (PPTA) under different strengths of El Niño and La Niña conditions. Based on the findings, Fig. 6 shows correlations between PPTA and the Niño indices for different severities of El Niño and La Niña (measured from Niño 3.4 such that  $\pm 0.5$  to  $\pm 1.0$  is weak,  $\pm 1.0$  to  $\pm 1.5$  is moderate and  $\pm 1.5$  above/below is strong) [5]. Niño 1 + 2 has a strong La Niña correlation (0.76), indicating a strong positive relationship. PPT increases more than usual as the Niño 1 + 2 index rises following the Strong La Niña phase. In contrast to Niño 1 + 2, since the correlation for Niño 3, 3.4, and 4 during Strong La Niña events is  $-0.64/-0.50/-0.65$ , it suggests a moderate to strong negative relationship. If the SSTa in the Pacific Ocean around Niño 3.4 is lower than  $-1.5$  °C, Thailand's PPTA will increase significantly more than usual. Since the SSTa in the Pacific Ocean (especially Niño 3, 3.4, and 4) are significantly cooler than average in strong La Niña, caused by the strengthening of the Walker Circulation. This results in stronger trade winds, enhanced upwelling of cold water in the eastern Pacific, and a shift in convection westward towards the western Pacific and Southeast Asia. However, the easternmost Niño 1 + 2 region can experience a warming or less intense cooling compared to the central Pacific. It could occur due to localized oceanic and atmospheric feedback, such as reduced upwelling or the influence of the South American coastal currents. The warming or reduced cooling in Niño 1 + 2 during a Strong La Niña can lead to a different atmospheric response, such as a local weakening of the eastern branch of the Walker Circulation. It can create conditions with a positive correlation between Niño 1 + 2 and PPTA. Regarding other severity (Strong/Moderate/Weak for El Niño and Moderate/Weak for La Niña), the correlation is low/moderate ( $-0.01$  to  $-0.32$  and  $0.01$  to  $0.29$ ). Additionally, these relationships are notable, showing that Strong La Niña in all Niño regions significantly impacts PPTA (direct impact from Niño 1 + 2, and opposite for Niño 3, 3.4, 4). These correlations highlight the periods in which the Niño indices have a more pronounced impact on PPTA. This understanding is crucial for the potential impacts of ENSO on PPTA.

#### Wavelet analysis

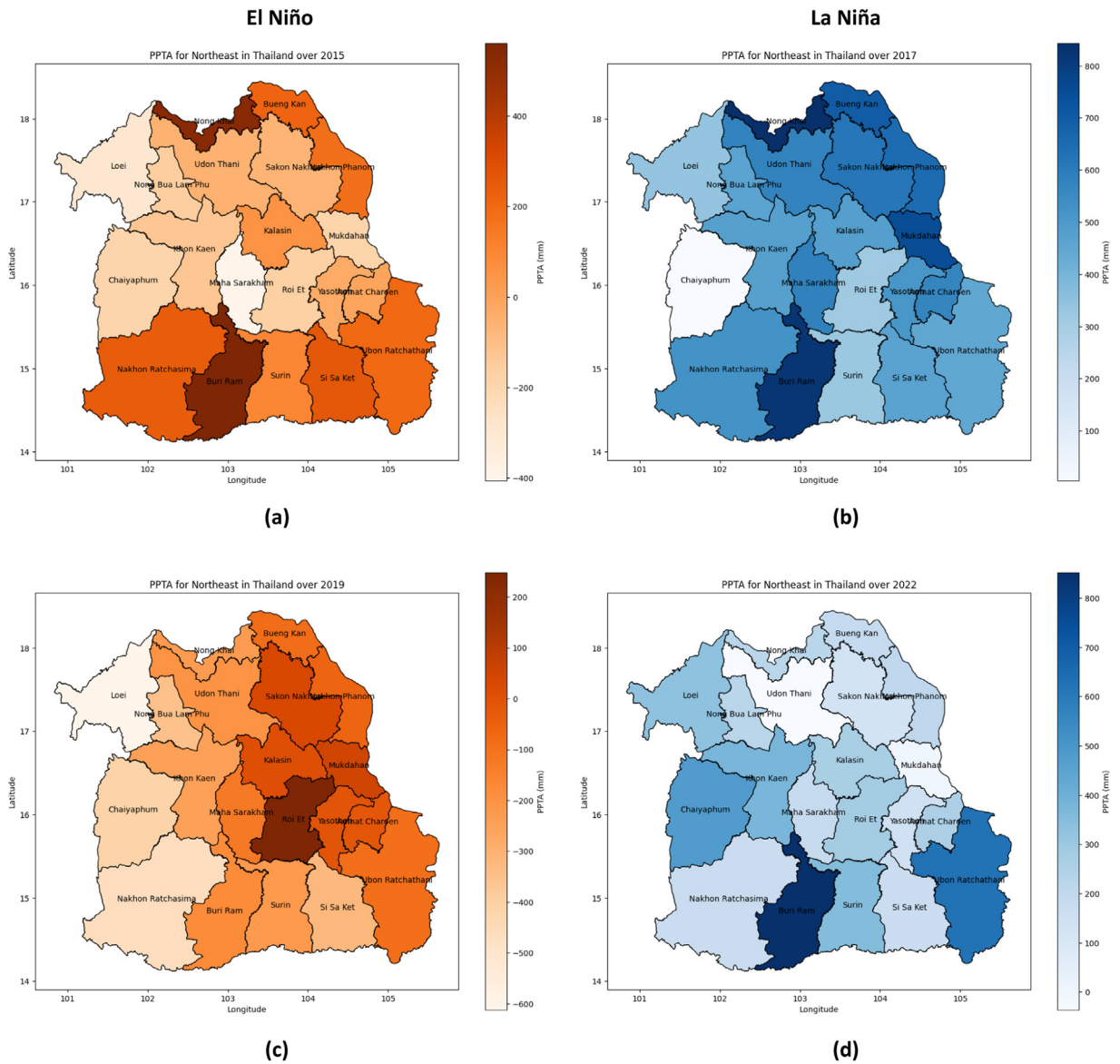
Wavelet coherence is a powerful method for examining the relationship between two-time series in both time and frequency domains. Fig. 7 shows the coherence between the Niño indices (1 + 2, 3, 3.4, 4) and PPTA in Northeast Thailand from 1993 to 2022. The Y-axis shows the relationship's periodicity (in years), ranging from 0.25 to 8 years. This color scale, ranging from blue (low coherence) to red (high coherence), indicates the strength of the relationship at different times and frequencies. Fig. 7(a) shows intermittent coherence periods in 1993–1998, 2008–2013, and 2013–2018. There was a significant coherence between approximately 1.5 to 4 years, with the relationship between the Niño 1 + 2 and PPTA being extremely opposite around this time. A substantial coherence is seen between 2003 and 2008 at a time interval of roughly 0.5 years, indicating a semi-annual link. Fig. 7(b) and (c) display stronger coherence than Niño 1 + 2, mainly between 2 and 4 years and 4 to 7 years in 1993–2003. It indicates that during this time, there was a relationship between the Niño 3–3.4 and PPTA in Northeast Thailand at these timescales. In 2003–2013, the plot showed notable coherence at periods around 0.5 to 1 year (semi-annual to annual cycles). Lastly, in 2013–2018, the coherence



**Fig. 7.** Wavelet coherence between Niño index (a) Niño 1 + 2, (b) Niño 3, (c) Niño 3.4, (d) Niño 4 and PPTA in Northeast Thailand. Blue denotes a weaker power, and dark red denotes a stronger power. Rightward arrows ( $\rightarrow$ ) indicate positively correlated. Leftward arrows ( $\leftarrow$ ) signify a negative relationship. The COI indicates a 95 % significance level in contrast to the noise.

is strong for around 2 to 4 years. The right-pointing arrows suggest that during this time, the Niño 3–3.4 and PPTA, indicating a strong correlation between these variables at these specific periods. While Fig. 7(d) shows around the years 1998–2000 at a period of about 2–4 years, there is coherence, suggesting that during this time, the Niño 4 and PPTA were strongly linked. The leftward arrows throughout the plot suggest that Niño 3–3.4 and PPTA typically change opposite. The coherence between PPTA and Niño indices throughout several periods implies an ocean-atmosphere mechanism. The SSTa in the Niño region affects PPTA in Northeast Thailand. These relationships are stronger during certain phases of ENSO, leading to the observed coherence. This is because ENSO events typically occur every 2 to 7 years, and the associated shifts in atmospheric circulation directly affect the monsoon and precipitation patterns in Northeast Thailand. Coherence at these short periods might be related to annual cycles, reflecting the influence of ENSO on specific seasons within the year. The coherence at specific time scales reflects large-scale ocean-atmosphere interactions that impact PPT variability in Northeast Thailand.

Fig. 8 shows the sample effect of El Niño and La Niña in Northeast Thailand. El Niño is a climate phenomenon characterized by warming sea surface temperatures in the Pacific Ocean. It influences weather patterns globally, including in Thailand. During El Niño years, such as 2015 and 2019. The PPTA decreased in almost all provinces. It means that the region experienced lower-than-mean rainfall. The reduction in rainfall often leads to drought conditions, affecting agriculture, water supply, and daily life. Farmers, in particular, struggle with crop yields, which can have economic impacts, leading to potential water shortages. Unlike La Niña events, it is characterized by cooler-than-mean sea surface temperatures in the Pacific Ocean. During La Niña years, such as 2017 and 2022, PPTA increased in almost all provinces, indicating higher-than-average PPT and the increased PPT affected by ENSO. Completely



**Fig. 8.** PPTA for each province in Northeast Thailand over El Niño years (a) 2015, (c) 2019 and La Niña years (b) 2017, (d) 2022.

differences between the El Niño and La Niña phases in the precipitation. While excessive rainfall can be detrimental, moderate increases benefit agricultural production by providing adequate water for crops.

### Discussion

The fluctuation of climate variability is influenced by several uncertainties, including ENSO [46]. The analysis of the ENSO impacts on precipitation variability in Northeast Thailand reveals complex interactions between climatic variables and ENSO phases. The study utilized a comprehensive dataset spanning 30 years from 1993 to 2022, employing Pearson's correlation and wavelet coherence analysis to elucidate these relationships. The findings indicate that La Niña events have a more pronounced effect on precipitation patterns in Northeast Thailand than El Niño events. During strong La Niña phases, precipitation tends to increase significantly, as evidenced by the negative correlation coefficients from Niño regions (Niño 3, 3.4, and 4), which were  $-0.64$ ,  $-0.50$ , and  $-0.65$ , respectively. This increase in precipitation can benefit agriculture by providing much-needed water during the wet season but also poses risks of flooding and related hazards. However, the peak in correlation indicates the time it takes for atmospheric changes (captured by SSTA) to impact PPTA significantly. It peaks around lag 4 to lag 6 before decreasing slightly. For example, the correlation in the Niño 4 region goes from  $-0.08$  at lag 0 to  $-0.16$  at lag 4 and then to  $-0.06$  at lag 12. Based on the findings,

each Niño region's effect on PPTA with different lag months is as follows: Niño 1, 3 is 5 months, Niño 3.4, 4 is 4 months with high correlations with PPTA.

Conversely, El Niño phases are associated with reduced rainfall in the region, which can exacerbate drought conditions. These dry conditions impact agricultural productivity and water availability and increase the risk of forest fires. Wavelet transform coherence (WTC) analysis further revealed the temporal and frequency-specific nature of the ENSO-climate relationship. This method allowed for the identification of periods where ENSO signals were more strongly correlated with precipitation patterns, providing the relationship of 2–5 years into the timing and intensity of ENSO impacts. Thai farmers will benefit from this study if it is enhanced by evaluating the possibility of expected effects on agricultural output [47]. We can predict yields by looking at when ENSO will impact agricultural output.

Furthermore, if we can forecast when it is extreme, we can develop better tactics to deal with it. Extended research duration may provide additional understanding of the relationship between ENSO and climate factors. Unfortunately, limited data on climate variables meant we could only look at this time frame.

## Conclusion

This study has demonstrated the significant impact of the ENSO on PPTA in Northeast Thailand. Through rigorous analysis using Pearson's correlation and wavelet coherency, it is evident that El Niño and La Niña events substantially influence rainfall distribution and intensity in the region. During El Niño years, Northeast Thailand experiences reduced rainfall, often leading to drought conditions that adversely affect agriculture, water supply, and daily life. It is contrasted by La Niña years, which bring increased rainfall and the risk of flooding yet also provide beneficial water for agricultural activities. These findings underscore the importance of understanding ENSO's variability and its implications for regional climate patterns. This knowledge can aid in better preparing for and mitigating the adverse effects of extreme weather events linked to ENSO, thus enhancing the resilience of local communities and economies. Future research should continue to explore the complexities of ENSO and its broader climatic impacts, incorporating more extensive datasets and advanced analytical methods to refine predictions and inform policy decisions. It will be crucial in addressing the challenges of climate variability and ensuring sustainable development in Northeast Thailand and similar regions globally.

## Limitations

The analysis relies on available meteorological data, which may have limitations in coverage, accuracy, and consistency over the study period. It could affect the precision of the correlation and wavelet coherency results. Additionally, the findings are specific to one region of Thailand.

## Ethics statements

The data used in this research are secondary data derived from the office of the Thai Meteorological Department (TMD) and National Oceanic and Atmospheric Administration (NOAA).

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgments

The first author was supported by Petchara Pra Jom Klao master's degree research scholarship from King Mongkut's University of Technology Thonburi (KMUTT), Thailand Science Research and Innovation (TSRI), and National Science, Research, and Innovation Fund (NSRF).

## Funding

No Funding was used in this research.

## References

- [1] K.E. Trenberth, El Niño southern oscillation (ENSO), *Sea* (1996).
- [2] R.W. Katz, Sir Gilbert Walker and a Connection between El Niño and Statistics, *Stat. Sci.* 17 (1) (2002) 97–112 16.
- [3] M.J. McPhaden, S.E. Zebiak, M.H. Glantz, ENSO as an integrating concept in earth science, *Science* 314 (5806) (2006) 1740–1745.
- [4] J.A. Amador, et al., Atmospheric forcing of the eastern tropical pacific: a review, *Prog. Oceanogr.* 69 (2–4) (2006) 101–142.
- [5] X. Jia, J. Ge, S. Wang, Diverse impacts of ENSO on wintertime rainfall over the maritime continent, *Int. J. Climatol.* 36 (9) (2016).



- [6] J. Lisonbee, J. Ribbe, Seasonal climate influences on the timing of the Australian monsoon onset, *Weather Clim. Dynam.* 2 (2) (2021) 489–506.
- [7] N. Singhrattana, et al., Interannual and interdecadal variability of Thailand summer monsoon season, *J. Clim.* 18 (11) (2005) 1697–1708.
- [8] A.M. Grimm, R.G. Tedeschi, ENSO and extreme rainfall events in South America, *J. Clim.* 22 (7) (2009) 1589–1609.
- [9] S. Kirtphaiaboon, et al., Rainfall variability over Thailand related to the El Niño-Southern Oscillation (ENSO), *J. Sustain. Energy Environ.* 5 (2) (2014) 37–42.
- [10] T.V. Vu, et al., Effects of ENSO on autumn rainfall in central Vietnam, *Adv. Meteorol.* 2015 (2015).
- [11] T.F. Loughran, A.J. Pitman, S.E. Perkins-Kirkpatrick, The El Niño–Southern Oscillation’s effect on summer heatwave development mechanisms in Australia, *Clim. Dyn.* 52 (9) (2019) 6279–6300.
- [12] A.N. Angelakis, et al., Evolution of floods: from ancient times to the present times (CA 7600 BC to the present) and the future, *Land* 12 (6) (2023) 1211.
- [13] R.H. Kripalani, A. Kulkarni, Rainfall variability over South-east Asia—Connections with Indian monsoon and ENSO extremes: new perspectives, *Int. J. Climatol.* 17 (11) (1997) 1155–1168.
- [14] C. Muangsong, et al., An annually laminated stalagmite record of the changes in Thailand monsoon rainfall over the past 387 years and its relationship to IOD and ENSO, *Quat. Int.* 349 (2014) 90–97.
- [15] P. Wongkhunkaew, S. Konyai, V. Sriboonlue, Effect of climate variability on rice production of Northeastern Thailand, *GEOMATE J.* 18 (68) (2020) 129–134.
- [16] S. Chawchai, et al., Hydroclimate variability of western Thailand during the last 1400 years, *Quat. Sci. Rev.* 241 (2020) 106423.
- [17] R. D’Arrigo, et al., On the variability of ENSO over the past six centuries, *Geophys. Res. Lett.* 32 (3) (2005).
- [18] M. Sifuzzaman, M.R. Islam, M.Z. Ali, Application of wavelet transform and its advantages compared to Fourier transform. 2009.
- [19] S. Jevrejeva, J. Moore, A. Grinsted, Influence of the Arctic Oscillation and El Niño-Southern Oscillation (ENSO) on ice conditions in the Baltic Sea: the wavelet approach, *J. Geophys. Res.: Atmos.* 108 (D21) (2003).
- [20] C. Torrence, G.P. Compo, A Practical Guide to Wavelet Analysis: with significance and confidence testing, Program in Atmospheric and Oceanic Sciences, University of Colorado at Boulder, 2011.
- [21] K.A. Tamaddun, A. Kalra, S. Ahmad, Wavelet analyses of western US streamflow with ENSO and PDO, *J. Water Clim. Change* 8 (1) (2017) 26–39.
- [22] D. Penny, A 40,000 year palynological record from north-east Thailand; implications for biogeography and palaeo-environmental reconstruction, *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 171 (3–4) (2001) 97–128.
- [23] M. Waqas, et al., Potential of artificial intelligence-based techniques for rainfall forecasting in thailand: a comprehensive review, *Water* 15 (16) (2023) 2979.
- [24] M.C. Peel, B.L. Finlayson, T.A. McMahon, Updated world map of the Köppen-Geiger climate classification, *Hydrol. Earth Syst. Sci.* 11 (5) (2007) 1633–1644.
- [25] O. Sabatini, The Agricultural Economy of Thailand, 321, US Department of Agriculture, Economic Research Service, 1972.
- [26] B. Huang, et al., Extended reconstructed sea surface temperature, version 5 (ERSSTv5): upgrades, validations, and intercomparisons, *J. Clim.* 30 (20) (2017) 8179–8205.
- [27] D.E. Hanley, et al., A quantitative evaluation of ENSO indices, *J. Clim.* 16 (8) (2003) 1249–1258.
- [28] K.E. Trenberth, D.P. Stepaniak, Indices of el Niño evolution, *J. Clim.* 14 (8) (2001) 1697–1701.
- [29] V. Misra, et al., Climate scenarios: a Florida-Centric view. 2011.
- [30] H. Ba, et al., Improving ANN model performance in runoff forecasting by adding soil moisture input and using data preprocessing techniques, *Hydrol. Res.* 49 (3) (2018) 744–760.
- [31] A. Wangwongchai, et al., Imputation of missing daily rainfall data; A comparison between artificial intelligence and statistical techniques, *MethodsX* 11 (2023) 102459.
- [32] M.H. Fiero, et al., Statistical analysis and handling of missing data in cluster randomized trials: a systematic review, *Trials* 17 (2016) 1–10.
- [33] M.J. Baxter, C.E. Buck, in: Data Handling and Statistical Analysis, Chemical Analysis-New York-Interscience Then John Wiley, 2000, pp. 681–746.
- [34] J.M. Shaughnessy, J. Garfield, B. Greer, Data handling, *Int. Handbook Math. Educ.* (1996) 205–237.
- [35] J. Adler, I. Parmryd, Quantifying colocalization by correlation: the Pearson correlation coefficient is superior to the Mander’s overlap coefficient, *Cytometry Part A* 77 (8) (2010) 733–742.
- [36] M. Farge, Wavelet transforms and their applications to turbulence, *Annu. Rev. Fluid Mech.* 24 (1) (1992) 395–458.
- [37] M. Shoaib, A.Y. Shamseldin, B.W. Melville, Comparative study of different wavelet based neural network models for rainfall–runoff modeling, *J. Hydrol.* 515 (2014) 47–58.
- [38] L. Chun-Lin, in: A Tutorial of the Wavelet Transform, 21, NTUEE, Taiwan, 2010, p. 2.
- [39] J.B. Tary, R.H. Herrera, M. Van Der Baan, Analysis of time-varying signals using continuous wavelet and synchrosqueezed transforms, *Philos. Trans. R. Soc., A* 376 (2126) (2018) 20170254.
- [40] K.A. Tamaddun, et al., Multi-scale correlation between the western US snow water equivalent and ENSO/PDO using wavelet analyses, *Water Resour. Manage.* 31 (2017) 2745–2759.
- [41] C. Torrence, P.J. Webster, Interdecadal changes in the ENSO–monsoon system, *J. Clim.* 12 (8) (1999) 2679–2690.
- [42] A. Grinsted, J.C. Moore, S. Jevrejeva, Application of the cross wavelet transform and wavelet coherence to geophysical time series, *Nonlinear Process. Geophys.* 11 (5/6) (2004) 561–566.
- [43] T.S. Kestin, et al., Time-frequency variability of ENSO and stochastic simulations, *J. Clim.* 11 (9) (1998) 2258–2272.
- [44] L. Aguiar-Conraria, N. Azevedo, M.J. Soares, Using wavelets to decompose the time–frequency effects of monetary policy, *Physica A* 387 (12) (2008) 2863–2878.
- [45] G. Luo, et al., Assessment of the characteristics and influencing factors of ozone in Fuzhou, China, using wavelet analysis, *Aerosol. Air Qual. Res.* 20 (9) (2020) 1898–1909.
- [46] A.E. Lawman, et al., A century of reduced ENSO variability during the medieval climate anomaly, *Paleoceanogr. Paleoclimatol.* 35 (4) (2020) e2019PA003742.
- [47] G. Podesta, et al., Use of ENSO-related climate information in agricultural decision making in Argentina: a pilot experience, *Agric. Syst.* 74 (3) (2002) 371–392.